F78PB Probability & Statistics B, April 2024/25 Solutions

Maps to CS1 Actuarial Statistics exemption.
Weighting within CS1 exemption: 25%.

1. *Parts (a) - (c) seen, parts (d)-(e) unseen but similar to seen examples*
CS1 syllabus objectives 1.2.7, 1.4.1, 1.4.3, 1.4.4

(a) Let $Z \sim \mathrm{N}(0,1)$ and set $X = \mu + \sigma Z$. By definition the moment generating function of X is given by

$$M_X(\theta) = \mathbb{E}[e^{\theta X}]$$

whenever this expectation exists. Using that $X = \mu + \sigma Z$ we have

$$M_X(\theta) = \mathbb{E}[e^{\mu\theta + \sigma\theta Z}]$$

[1 mark]

Rearranging this gives

$$M_X(\theta) = e^{\mu\theta} M_Z(\sigma\theta).$$

[1 mark]

Now using the expression for the moment generating function of a standard normal random variable given in the question or in the formula sheet

$$M_X(\theta) = e^{\mu\theta} e^{\frac{1}{2}\sigma^2\theta^2} = e^{\mu\theta + \frac{1}{2}\sigma^2\theta^2}.$$

[1 mark]

(b) If $\mu = 1$ and $\sigma = 2$ then the moment generating function is given by

$$M_X(\theta) = e^{\theta} e^{2\theta^2}.$$

Using the exponential series

$$\begin{aligned} e^{\theta} &= 1 + \theta + \frac{1}{2}\theta^2 + \frac{1}{3!}\theta^3 + \ldots \\ e^{2\theta^2} &= 1 + 2\theta^2 + \ldots \end{aligned}$$

[1 mark]

Here we have truncated both series after θ^3 since higher order terms will not affect the remaining calculation. Multiplying the series together

$$M_X(\theta) = \left(1 + \theta + \frac{1}{2}\theta^2 + \frac{1}{3!}\theta^3 + \ldots\right)\left(1 + 2\theta^2 + \ldots\right)$$

[1 mark]

Alternatively, one can use the exponential series for $e^{\theta + 2\theta^2}$ to obtain

$$\begin{aligned} M_X(\theta) = e^{\theta + 2\theta^2} &= 1 + (\theta + 2\theta^2) + \frac{1}{2}(\theta + 2\theta^2)^2 + \frac{1}{6}(\theta + 2\theta^2)^3 + \ldots \\ &= 1 + (\theta + 2\theta^2) + (\frac{1}{2}\theta^2 + 2\theta^3 + 2\theta^4) + \frac{1}{6}\theta^3 + \ldots \end{aligned}$$

Here we have truncated the series to not include an terms of order θ^4 or higher. [using this approach also receives 2 marks for obtaining the correct expansion] Now collecting order θ^3 terms we have the following:

$$\frac{1}{3!}\theta^3 + 2\theta^3 = \frac{13}{6}\theta^3$$

[2 marks]

In general, the moment generating function has the power series

$$M_X(\theta) = \sum_{k=0}^{\infty} \frac{\mathbb{E}[X^k]}{k!}\theta^k.$$

Equating order θ^3 terms gives

$$\frac{\mathbb{E}[X^3]}{3!} = \frac{13}{6} \Rightarrow \mathbb{E}[X^3] = 13.$$

[1 mark]

[Alternative calculation: If the third moment is found by differentiating the MGF then 5 marks are still awarded]

(c) Let $Z_1, \ldots, Z_n$ be a sequence of independent standard normal random variables, let

$$S = Z_1^2 + Z_2^2 + \ldots + Z_n^2.$$

Since $Z_1, \ldots, Z_n$ are i.i.d. we have

$$M_S(\theta) = M_{Z^2}(\theta)^n.$$

[1 mark]

Using the answer to part (c) we have

$$M_S(\theta) = (1-2\theta)^{-\frac{n}{2}} = \left(\frac{1/2}{1/2-\theta}\right)^{\frac{n}{2}}.$$

for $\theta < 1/2$. [1 mark]

The moment generating function for S coincides with the moment generating function for a χ^2-distribution with n degrees of freedom or equivalently a Gamma$(\frac{n}{2}, \frac{1}{2})$-distribution. Therefore by the uniqueness property of the Moment generating function

$$S \sim \chi^2_n \text{ Or } S \sim \text{Gamma}(\frac{n}{2}, \frac{1}{2}).$$

[1 mark for stating correct distribution]

[Total 11 marks]

2. *All parts unseen but similar to examples*
CS1 syllabus objectives 3.1.3, 3.1.4, 3.1.5

(a) The bias for $\hat{\beta}_1$ is given by:

$$\begin{aligned}\text{Bias}(\hat{\beta}_1) &= \mathbb{E}[\hat{\beta}_1] - \beta \\ &= \mathbb{E}\left[\frac{4\bar{X}}{3}\right] - \beta \\ &= \frac{4}{3}\mathbb{E}[\bar{X}] - \beta \\ &= \frac{4}{3} \cdot \frac{3\beta}{4} - \beta \\ &= 0\end{aligned}$$

Thus, $\hat{\beta}_1$ is an unbiased estimator of β.

[2 marks]

(b) The MSE of $\hat{\beta}_1$ is given by:

$$\begin{aligned}\text{MSE}(\hat{\beta}_1) &= \mathbb{V}\text{ar}(\hat{\beta}_1) + \text{Bias}(\hat{\beta}_1)^2 \\ &= \mathbb{V}\text{ar}\left(\frac{4\bar{X}}{3}\right) + 0 \\ &= \frac{16}{9n}\mathbb{V}\text{ar}(X) \\ &= \frac{16}{9n} \cdot \frac{3\beta^2}{80} \\ &= \frac{\beta^2}{15n}\end{aligned}$$

[2 marks]

(c) The MSE of $\hat{\beta}_2$ is given by:

$$\text{MSE}(\hat{\beta}_2) = \mathbb{V}\text{ar}(\hat{\beta}_2) + \text{Bias}(\hat{\beta}_2)^2$$

$$\mathbb{V}\text{ar}\left(\hat{\beta}_2\right) = \gamma^2\mathbb{V}\text{ar}(\bar{x}) = \gamma^2 \cdot \frac{\mathbb{V}\text{ar}(x)}{n} = \gamma^2 \cdot \frac{3\beta^2}{80n}$$

[1 mark]

$$\mathbb{E}\left(\hat{\beta}_2\right) = \gamma\mathbb{E}(\bar{x}) = \gamma \cdot \frac{3\beta}{4}$$

$$\begin{aligned}\text{MSE}\left(\hat{\beta}_2\right) &= \gamma^2 \cdot \frac{3\beta^2}{80n} + \left(\gamma \cdot \frac{3\beta}{4} - \beta\right)^2 \\ &= \gamma^2 \cdot \frac{3\beta^2}{80n} + \beta^2\left(\frac{3\gamma}{4} - 1\right)^2\end{aligned}$$

[1 mark]

(d) To minimize the MSE, we take the derivative of the MSE expression with respect to γ and set it equal to zero:

$$\frac{d}{d\gamma}\left[\text{MSE}\left(\hat{\beta}_2\right)\right] = 0$$

Simplify the MSE as:

$$\text{MSE}\left(\hat{\beta}_2\right) = \frac{3\beta^2}{80n}\gamma^2 + \frac{9\beta^2}{16}\gamma^2 - \frac{3\beta^2}{2}\gamma + \beta^2$$

$$\begin{aligned}\frac{d}{d\gamma}\left[\text{MSE}\left(\hat{\beta}_2\right)\right] &= \frac{6\beta^2}{80n}\gamma + \frac{18\beta^2}{16}\gamma - \frac{3\beta^2}{2} \\ &= \frac{3\beta^2}{40n}\gamma + \frac{9\beta^2}{8}\gamma - \frac{3\beta^2}{2}\end{aligned}$$

[1 mark]

$$\frac{3\beta^2}{40n}\gamma + \frac{9\beta^2}{8}\gamma - \frac{3\beta^2}{2} = 0$$

$$\frac{3}{40n}\gamma + \frac{9}{8}\gamma = \frac{3}{2}$$

$$\gamma\left(\frac{3}{40n} + \frac{9}{8}\right) = \frac{3}{2}$$

$$\gamma\left(\frac{3+45n}{40n}\right) = \frac{3}{2}$$

$$\gamma = \left(\frac{60n}{3+45n}\right)$$

[2 mark]

The most efficient estimator of the form $\hat{\beta}_2 = \gamma\bar{x}$ is given by:

$$\hat{\beta}_2^{\,*} = \frac{60n}{3+45n}\bar{x}$$

[1 mark]

[Total 10 marks]

3. *All parts unseen but similar to examples*
CS1 syllabus objectives 3.1.1,3.1.2, 3.1.3, 3.1.4, 3.1.5

(a) The maximum likelihood estimator (MLE) for θ:
The likelihood function is given by:

$$L(\theta) = \prod_{i=1}^{n} f(x_i) = \prod_{i=1}^{n} \left(\frac{1}{\theta} x_i^{(1-\theta)/\theta} \right)$$
$$= \frac{1}{\theta^n} \prod_{i=1}^{n} x_i^{(1-\theta)/\theta}$$

[1 mark]

Taking the natural logarithm to get the log-likelihood:

$$\ell(\theta) = \log L(\theta) = -n \log \theta + \frac{(1-\theta)}{\theta} \sum_{i=1}^{n} \log x_i$$

[1 mark]

Differentiate $\ell(\theta)$ with respect to θ and set it equal to zero to find the maximum likelihood estimator (MLE) for θ:

$$\frac{d\ell(\theta)}{d\theta} = -\frac{n}{\theta} - \frac{1}{\theta^2} \sum_{i=1}^{n} \log x_i$$

$$-\frac{n}{\theta} - \frac{1}{\theta^2} \sum_{i=1}^{n} \log x_i = 0$$

[1 mark]

Solving for θ :

$$\frac{n}{\theta} = -\frac{1}{\theta^2} \sum_{i=1}^{n} \log x_i$$
$$\theta^* = -\frac{\sum_{i=1}^{n} \log x_i}{n}$$

[1 mark]

It remains to verify that this is indeed the maximum value, for this we use the second derivative test (or a suitable plot)

$$\frac{d^2\ell}{d\lambda^2}(\theta) = \frac{n}{\theta^2} + \frac{2}{\theta^3} \sum_{i=1}^{n} \log(x_i) = \frac{n}{\theta^3} \left(\theta + \frac{2}{n} \sum_{i=1}^{n} \log(x_i) \right).$$

Evaluating this at the critical point we have

$$\frac{d^2\ell}{d\lambda^2}(\theta^*) = \frac{n}{(\theta^*)^3} \left(\frac{1}{n} \sum_{i=1}^{n} \log(x_i) \right).$$

Since this is negative (as $x_i < 1$) we see that θ^* is the maximum and hence $\theta_{\text{MLE}} = \theta^*$.

[1 mark]

(b) The method of moment estimator (MME) for θ:
Equate the population moment to the sample moment:

$$\mathbb{E}(X) = \mu_1$$

$$\begin{aligned}\mathbb{E}[X] = \int_0^1 xf(x)dx &= \int_0^1 x\left(\frac{1}{\theta}x^{(1-\theta)/\theta}\right)dx \\ &= \frac{\mathbf{1}}{\theta}\int_0^1 x^{\frac{1}{\theta}}dx \\ &= \frac{\mathbf{1}}{\theta}\left[\frac{x^{\frac{1+\theta}{\theta}}}{\frac{1+\theta}{\theta}}\right]_0^1 = \frac{1}{1+\theta}\end{aligned}$$

[2 marks]

$$\mu_1 = \frac{1}{n}\sum_{i=1}^{n} x_i = \bar{x}$$

Thus, we set:

$$\bar{x} = \frac{1}{1+\theta}$$

[1 mark]

Now, solve for θ :

$$\begin{aligned}\bar{x}(1+\theta) &= 1 \\ \bar{x} + \bar{x}\theta &= 1 \\ \bar{x}\theta &= 1 - \bar{x} \\ \hat{\theta}_{MME} &= \frac{1-\bar{x}}{\bar{x}}\end{aligned}$$

[1 mark]

(c) Comparison of the estimators based on the data:

$$\hat{\theta}_{MLE} = -\frac{\sum_{i=1}^{n}\log x_i}{n} = -\frac{\log(0.11)+\log(0.22)+\log(0.32)}{3} \approx 0.741$$

[1 mark]

$$\hat{\theta}_{MME} = \frac{1-\bar{x}}{\bar{x}} = \frac{1-0.22}{0.22} \approx 3.545$$

Thus, the MLE provides the smallest estimate for θ.

[1 mark]

[Total 11 marks]

4. *First part bookwork, all other parts unseen but similar to examples*
CS1 syllabus objectives 3.2.1, 3.2.3, 3.2.4, 3.2.6

(a) i. The quantity

$$\frac{\bar{D} - \mu_D}{\sqrt{S_D^2/n}}$$

is a pivotal quantity for $\bar{D}$ because:

A. It is a function which involves the parameter μ_D and sample.

[1 mark]

B. It has a known distribution which does not depend on the parameter μ_d. That is, the quantity follows a t-distribution with $n-1$ degrees of freedom.

[1 mark, either statement is valid for the mark]

C. It is a monotonic function of the parameter μ_D.

[1 mark]

ii. The 95% confidence interval for μ_D is given by

$$\begin{aligned}
&\left(\bar{D} - t_{n-1,0.025}\sqrt{\frac{s_D^2}{n}}, \bar{D} + t_{n-1,0.025}\sqrt{\frac{s_D^2}{n}}\right) \\
&= \left(7 - t_{11,0.025}\sqrt{\frac{18}{12}}, 7 + t_{11,0.025}\sqrt{\frac{18}{12}}\right) \\
&= \left(7 - 2.201\sqrt{\frac{18}{12}}, 7 + 2.201\sqrt{\frac{18}{12}}\right) \\
&= (4.304, 9.696)\,.
\end{aligned}$$

[2 marks]

The 95% confidence interval $(4.304, 9.696)$ does not contain zero, which indicates that the difference in mean performance scores after and before the training programme is statistically significant.

[1 mark]

(b) For a 95% confidence interval for $\frac{\sigma_X^2}{\sigma_Y^2}$, we need the critical values from the F-distribution:

$$\begin{aligned}
F_{\alpha/2,n_X-1,n_Y-1} &= F_{0.025,8,7} = 4.899 \\
F_{\alpha/2,n_Y-1,n_X-1} &= F_{0.025,7,8} = 4.529
\end{aligned}$$

[1 mark]

The 95% confidence interval for $\frac{\sigma_X^2}{\sigma_Y^2}$, the ratio of the population variances of

the two materials, is given by:

$$\begin{aligned}
&\left(\frac{s_X^2}{s_Y^2}\frac{1}{F_{\alpha/2,n_X-1,n_Y-1}}, \frac{s_X^2}{s_Y^2}\frac{1}{F_{\alpha/2,n_Y-1,n_X-1}}\right) \\
&= \left(\frac{25}{40}\frac{1}{4.899}, \frac{25}{40}4.529\right) \\
&= (0.128, 2.831)\,.
\end{aligned}$$

[2 marks]

The interval $(0.128, 2.831)$ contains one. Therefore, the company may conclude that the variances of the strength measurements for the two materials do not differ significantly at 95% confidence level.

[1 mark]

[Total 10 marks]

5. *All parts unseen but similar to examples*
CS1 syllabus objectives 3.3.1, 3.3.2

(a) i. Independence of Samples: The two groups (Group A and Group B) must be independent of each other.

ii. Normality: The data from each group should be approximately normally distributed.

iii. Equal Population Variances: The two groups should have approximately equal variances.

[3 marks]

(b) The null and alternative hypotheses are given by:

$$H_0 : \mu_1 - \mu_2 = 0 \quad \text{vs} \quad H_1 : \mu_1 - \mu_2 < 0$$

[1 mark]

The pooled variance is given by:

$$s_p^2 = \frac{(n_1-1)s_1^2 + (n_2-1)s_2^2}{n_1+n_2-2} = \frac{(10-1)7^2 + (12-1)6^2}{10+12-2} = 41.85.$$

[2 marks]

Then the test statistic is given by:

$$t = \frac{\bar{x}_1 - \bar{x}_2}{\sqrt{s_p^2\left(\frac{1}{n_1}+\frac{1}{n_2}\right)}} = \frac{75-80}{\sqrt{41.85\left(\frac{1}{10}+\frac{1}{12}\right)}} \approx -1.8.$$

[2 marks]

(c) The number of degrees of freedom for the t distribution: $n_1 + n_2 - 2 = 10 + 12 - 2 = 20$. The p-value is

$$\mathbb{P}(t < -1.8) = \mathbb{P}(t > 1.8) = 1 - \mathbb{P}(t < 1.8) = 1 - 0.9565 = 0.0435$$

[1 mark]

Since the p-value is 0.0435, we have some evidence to reject H_0. This means there is sufficient evidence to conclude that the average test score for Group B (innovative method) is higher than that of Group A (traditional method). [1 mark]

[Total 10 marks]

6. *All parts unseen but similar to examples*
CS1 syllabus objectives 4.1.2, 4.1.3, 4.1.4

(a) The sample correlation coefficient is given by

$$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}}.$$

From the data we are given we find

$$S_{xx} = \sum x_i^2 - \frac{1}{n}\left(\sum x_i\right)^2 = 350 - \frac{58^2}{15} = 125.733$$
$$S_{yy} = \sum y_i^2 - \frac{1}{n}\left(\sum y_i\right)^2 = 3300 - \frac{210^2}{15} = 360$$
$$S_{xy} = \sum x_i y_i - \frac{1}{n}\left(\sum x_i\right)\left(\sum y_i\right) = 830 - \frac{(58)(210)}{15} = 18$$

This gives

$$r = \frac{S_{xy}}{\sqrt{S_{xx}S_{yy}}} = \frac{18}{\sqrt{(360)(125.733)}} = 0.0846.$$

[3 marks]

This shows there is no linear relationship between the amount spent on newspaper advertising and the sales figures.

[1 mark]

(b) We will test the hypotheses

$$H_0 : \rho = 0 \text{ vs } H_1 : \rho \neq 0.$$

The test statistic is

$$T = \frac{R\sqrt{n-2}}{\sqrt{1-R^2}}$$

where $R = r(\underline{Y}, \underline{Z})$ is the correlation coefficient of the random sample $(\underline{Y}, \underline{Z})$.

[1 mark for correct set up]

The observed value of the test statistic is

$$t = \frac{0.0846\sqrt{13}}{\sqrt{1 - 0.0846^2}} = 0.306$$

[1 mark]

The test statistic T is distributed according to t_{13} so 5% critical region is given by $(-\infty, -t_{13,\alpha/2}) \cup (t_{13,0.025}, \infty)$ and from the statistical tables

$$t_{13,0.025} = 2.160.$$

Hence the critical region is $(-\infty, -2.160) \cup (2.160, \infty)$.

[1 mark]

Since the observed value of the test statistic does not lie in the critical region we can not reject the null hypothesis. That is, there is not sufficient evidence to reject the claim that the correlation coefficient is significantly different from 0.

[1 mark for correct conclusion]

[Total 8 marks]